

Naive Bayes*(find and work with a partner)*

Say we have two tests for a specific disease. Each test (features f_1, f_2) can come back either positive “pos” or negative “neg”, and the true underlying condition of the patient is represented by y ($y = 1$ is “healthy” and $y = 2$ is “disease”). We observe this training data where $n = 7$ and $p = 2$:

$\mathbf{x}$	f_1	f_2	y
$\mathbf{x}_1$	pos	neg	1
$\mathbf{x}_2$	pos	pos	2
$\mathbf{x}_3$	pos	neg	2
$\mathbf{x}_4$	neg	neg	1
$\mathbf{x}_5$	pos	neg	2
$\mathbf{x}_6$	neg	neg	1
$\mathbf{x}_7$	neg	pos	2

1. To estimate the probability $p(y = k)$, for $k = 1, 2, \dots, K$, we will use the formula:

$$\theta_k = \frac{N_k + 1}{n + K},$$

where N_k is the count (“Number”) of data points where $y = k$. Compute θ_1 and θ_2 . What would θ_1 and θ_2 be if we in fact had *no* training data?

2. To estimate the probabilities $p(x_j = v \mid y = k)$ for all features j , values v , and class labels k , we will use the formula:

$$\theta_{k,j,v} = \frac{N_{k,j,v} + 1}{N_k + |f_j|},$$

where $N_{k,j,v}$ is the count of data points where $y = k$ and $x_j = v$, and $|f_j|$ is the number of possible values that f_j (feature j) can take on. Fill in the following tables with these θ values.

$y = 1$	pos	neg
f_1		
f_2		

$y = 2$	pos	neg
f_1		
f_2		

3. Continuing the example above, say we have a new data point $\mathbf{x}_{\text{test}} = [\text{neg}, \text{pos}]$. Our goal is to predict the class label based on the Naive Bayes posterior probability. In practice, we will compute this probability for each class k , based on our estimates (θ_k and $\theta_{k,j,v}$ terms). Then we will assign this data point the class label with maximum probability:

$$\hat{y} = \arg \max_{k \in \{1, 2, \dots, K\}} p(y = k \mid \mathbf{x}) = \arg \max_{k \in \{1, 2, \dots, K\}} p(y = k) \prod_{j=1}^p p(x_j \mid y = k).$$

For this $\mathbf{x}_{\text{test}}$, compute $p(y = 1 \mid \mathbf{x}_{\text{test}})$ and $p(y = 2 \mid \mathbf{x}_{\text{test}})$ and then assign a prediction label $\hat{y}$.

4. For the tennis example below, fill in the $\theta_{k,j,v}$ terms (thinking about how this could be implemented using dictionaries).

Day	Outlook	Temperature	Humidity	Wind	PlayTennis (y)
$\mathbf{x}_1$	Sunny	Hot	High	Weak	No
$\mathbf{x}_2$	Sunny	Hot	High	Strong	No
$\mathbf{x}_3$	Overcast	Hot	High	Weak	Yes
$\mathbf{x}_4$	Rain	Mild	High	Weak	Yes
$\mathbf{x}_5$	Rain	Cool	Normal	Weak	Yes
$\mathbf{x}_6$	Rain	Cool	Normal	Strong	No
$\mathbf{x}_7$	Overcast	Cool	Normal	Strong	Yes
$\mathbf{x}_8$	Sunny	Mild	High	Weak	No
$\mathbf{x}_9$	Sunny	Cool	Normal	Weak	Yes
$\mathbf{x}_{10}$	Rain	Mild	Normal	Weak	Yes
$\mathbf{x}_{11}$	Sunny	Mild	Normal	Strong	Yes
$\mathbf{x}_{12}$	Overcast	Mild	High	Strong	Yes
$\mathbf{x}_{13}$	Overcast	Hot	Normal	Weak	Yes
$\mathbf{x}_{14}$	Rain	Mild	High	Strong	No

y=No (0)

outlook	Sunny:	Overcast:	Rain:
temperature	Cool:	Mild:	Hot:
humidity	Normal:	High:	
wind	Weak:	Strong:	

y=Yes (1)

outlook	Sunny:	Overcast:	Rain:
temperature	Cool:	Mild:	Hot:
humidity	Normal:	High:	
wind	Weak:	Strong:	

Entropy and Information Gain*(find and work with a partner)*

Given the movie data below (Liked (Li) is the response variable), select the feature that maximizes the information gain. The steps below will guide you through the process.

Movie	Type	Length	Director	Famous actors	Liked?
m1	Comedy	Short	Adamson	No	Yes
m2	Animated	Short	Lasseter	No	No
m3	Drama	Medium	Adamson	No	Yes
m4	Animated	Long	Lasseter	Yes	No
m5	Comedy	Long	Lasseter	Yes	No
m6	Drama	Medium	Singer	Yes	Yes
m7	Animated	Short	Singer	No	Yes
m8	Comedy	Long	Adamson	Yes	Yes
m9	Drama	Medium	Lasseter	No	Yes

1. First compute the probability that the outcome is “yes”:

$$P(\text{Li} = \text{yes}) =$$

2. Next compute $H(\text{Li})$ using a calculator.

$$H(\text{Li}) =$$

3. Finally, given the conditional entropy values below, compute the information gain of each feature.

$$H(\text{Li} \mid \text{T}) = 0.61$$

$$H(\text{Li} \mid \text{Le}) = 0.61$$

$$H(\text{Li} \mid \text{D}) = 0.36$$

$$H(\text{Li} \mid \text{F}) = 0.85$$

$$\text{Gain}(\text{Li}, \text{T}) =$$

$$\text{Gain}(\text{Li}, \text{Le}) =$$